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Clinical and radiographic efficacy of photobiomodulation, platelet-rich fibrin, and their combination for socket preservation: a randomized clinical trial

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Abstract

Background The clinical and radiographic outcomes of platelet-rich fibrin (PRF), photobiomodulation treatment (PBM), and their combination for socket preservation were examined in this research.

Methods This single-blind randomized clinical trial assessed 72 individuals who had atraumatic extraction of anterior and premolar teeth. The patients were randomly allocated into four groups ($n = 18$): (I) PBM (660 nm, 100 mW, 2 J/cm^2, 10 s for soft tissue; 970 nm, 200 mW, 6 J/cm^2, 20 s for hard tissue in continuous-wave mode), administered once every other day for 2 weeks; (II) socket preservation with PRF; (III) PRF combined with PBM; and (IV) control (no intervention). Pain after 24, 48, and 72 h, soft tissue healing, keratinized gingiva width (KGW), and the buccolingual ridge width and height (at baseline, and 2 months post-extraction) were assessed clinically and radiographically. Data were analyzed using SPSS (alpha = 0.05).

Results Pain scores were significantly lower in the intervention groups than in the control group at all time points ($P < 0.05$). The PRF-PBM group showed lower pain scores than the PBM and PRF groups, although the difference was statistically significant only compared with the PBM group ($P < 0.05$). In the GEE analysis of soft tissue healing, PBM and PRF-PBM showed significantly better healing scores than the control group, whereas PRF alone did not differ significantly from the control group. In the ANCOVA analysis adjusted for baseline ridge width, all intervention groups showed significantly greater follow-up buccolingual ridge width than the control group, with the largest adjusted effect observed in the PRF-PBM group. No significant intergroup differences were observed for KGW or ridge height.

Conclusion The results showed that PRF-PBM may provide an early biological advantage for socket healing, particularly in reducing postoperative pain, improving early soft tissue healing, and preserving buccolingual ridge width. However, KGW and vertical ridge height were not significantly affected, and longer-term studies are needed to confirm whether these early benefits are maintained.

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Trial registration The study was registered in the clinical trial registry with the number IRCT20230222057497N1on 09.05.2023 and IRCT20230305057619N1 on 13.05.2023.

Keywords Platelet rich fibrin, Photobiomodulation, Low level laser therapy, Socket preservation

Introduction

Adequate height and width of the alveolar ridge are imperative for ideal implant placement and achieving optimal esthetics, stability, and function. However, the alveolar ridge width decreases following tooth extraction as a result of natural remodeling of alveolar bone, leading to severe bone resorption. This process would result in poor bone quality and quantity for implant placement, inappropriate crown/implant ratio, and esthetic problems, which are particularly concerning in the anterior region [1].

Socket preservation is defined as a conservative approach designed to preserve bone volume after tooth extraction [2], which include the application of various bone grafting materials and growth factors, with or without a resorbable or non-resorbable membrane [3–6]. Graft application can stimulate osteoblastic activity and lead to bone formation; however, grafts increase the risk of infection and have a gradual and slow degradation process, which can negatively impact soft and hard tissue healing in the extraction socket [5]. The graft replacement process with newly formed bone can be lengthy, and some residual graft material may remain at the site. Moreover, this method requires a high level of expertise and should be performed by experienced surgeons [7].

Photobiomodulation (PBM), also known as low-level laser therapy (LLLT) provides a source of optical energy for direct biological stimulation of cells [8]. It can regulate and enhance the healing process, decrease pain, increase tissue angiogenesis, and Adenosine Triphosphate (ATP) production in the mitochondria, and exert bio-stimulatory effects, which are known as photobiomodulation (PBM) [9]. PBM enhances bone regeneration by increasing the proliferation and differentiation of osteoblasts, calcium deposition, formation of type I collagen, ATP synthesis, and release of growth factors [10]. Moreover, PBM can eliminate or neutralize toxic compounds, such as bacterial endotoxins (lipopolysaccharides), which may enhance wound healing, and offer several advantages compared with the conventional mechanical treatment [11].

Platelet-rich fibrin (PRF) is a 100% autologous material that is composed of platelets, leukocytes, and cytokines within a robust fibrin network. It contains high levels of growth factors that are progressively released over a period of 7 to 14 days, thereby modulating the inflammatory phase of healing, promoting tissue regeneration, angiogenesis, and neovascularization, and reducing postoperative pain and edema [12–15]. The epithelial closure of surgical incisions is also expedited [12]. Additionally, PRF has the potential to reduce inflammation in periodontal tissue, speed up bone regeneration, improve the recovery of alveolar bone defects, and preserve the alveolar ridge [16]. PRF comprises many growth factors, including platelet-derived growth factor, transforming growth factor-beta, insulin-like growth factor, endothelial growth factor, fibroblast growth factor, and bone morphogenetic proteins. Moreover, as it is sourced from the patient's own blood, PRF does not need anticoagulants, hence reducing the chance of rejection [5]. The tissue engineering triangle consists of cells, scaffolds, and biochemical growth factors like PRF, which may improve therapeutic efficacy. Recent research reveals a fourth dimension, represented by photobiomodulation (PBM), which may expedite the healing process [17]. Despite promising preliminary evidence, clinical studies evaluating the efficacy of PBM and PRF, both individually and in combination, for regenerating human bone defects are limited in number. Existing studies primarily assessed bone quality via clinical examination or simple histological or radiographic analysis, lacking a comprehensive approach. Therefore, this study aimed to address the existing gap by assessing and comparing the clinical and radiographic effects of PBM, PRF, and their combination for socket preservation.

Materials & methods

Study design

This study was conducted at the Periodontology Department of the School of Dentistry, Mazandaran University of Medical Sciences.

A single-blind clinical trial was designed, with socket preservation performed using PBM in group 1, PRF application in extraction socket in group 2, PRF + PBM in group 3, and no intervention in the control group (group 4). The results were reported according to the Consolidated Standards of Reporting Trials (CONSORT) guidelines (Fig. 1) [18].

The present study was derived from two related thesis projects conducted at Mazandaran University of Medical Sciences. Each thesis project received separate ethical approval and was registered independently in the Iranian Registry of Clinical Trials before participant recruitment. The first approval and registration were IR.MAZUMS.REC.1401.497 and IRCT20230305057619N1, and the second were IR.MAZUMS.REC.1401.427 and IRCT20230222057497N1. Because the two thesis projects used the same clinical setting, eligibility criteria,

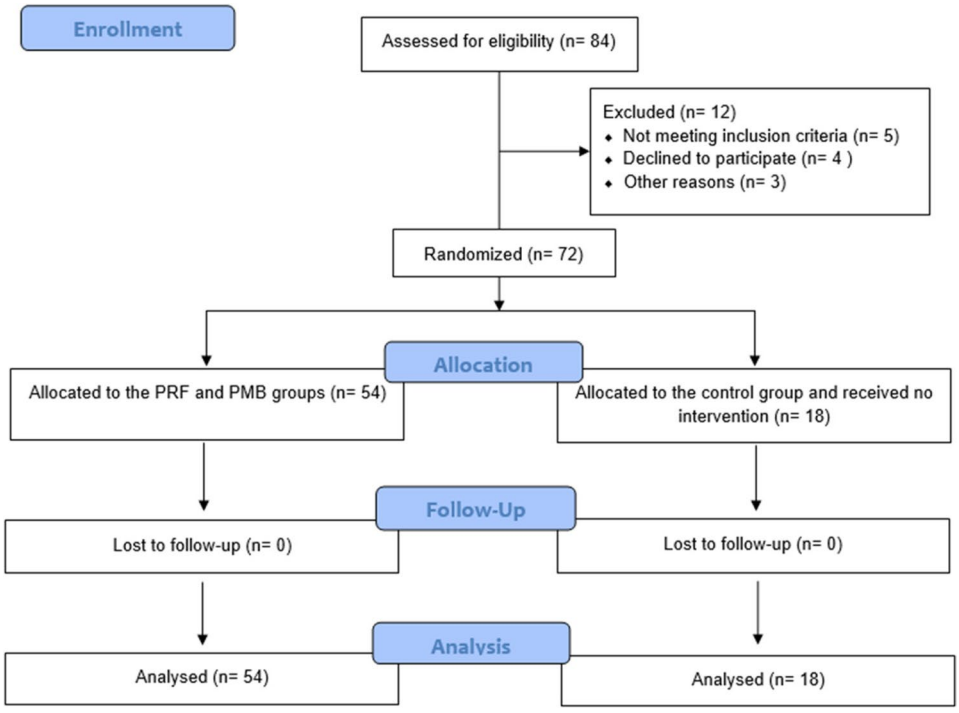


Fig. 1 CONSORT flow diagram of patient selection and allocation

extraction protocol, outcome assessments, follow-up schedule, and measurement methods, the study groups were combined into a single four-arm randomized clinical trial analysis comparing PBM, PRF, PRF-PBM, and control groups. All participants provided written informed consent before enrollment.

Patients

The subjects were chosen from a group of patients who had been referred for the extraction of anterior or premolar teeth in the mandible or maxilla. Patients were included if they were (I) older than 18 years and (II) required extraction of maxillary or mandibular anterior or premolar teeth because of irreparability, failed endodontic treatment, periodontal disease, or tooth fracture [1, 19, 20]; (III) had intact alveolar bone (defined as residual bone equal to at least two-thirds of the root length) [21]; (IV) had an extraction socket with four intact walls [22]; (V) were classified as ASA I or II [21]; (VI) agreed to participate and attend follow-up sessions; and did not require surgical extraction or use of a surgical bur.

The exclusion criteria were (I) patients with poor oral hygiene [1], (II) pregnancy or nursing [1], (III) heavy smoking (smoking more than 10 cigarettes/day) [23] or alcohol addiction [1], (IV) uncontrolled systemic diseases, history of head and neck radiotherapy, or immunocompromised state [24], (V) chronic intake of medications affecting bone metabolism and function, such as heparin, cyclosporin, bis-phosphonate, or chemotherapy medications [24, 25], (VI) the presence of noticeable periapical or periodontal lesions around the respective tooth, or the presence of dehiscence or fenestration in buccal bone plate [1, 19, 20, 26], (VII) contraindications for tooth extraction [27], and (VIII) acute infection or any other factor causing mouth-opening limitation [20].

Preoperative panoramic radiographs, together with clinical examination, were used for initial screening and assessment of eligibility, including evaluation of residual bone support and exclusion of periapical or periodontal pathology around the target tooth.

Sample size

The initial sample size calculation was performed based on buccolingual alveolar ridge width, which was considered the main radiographic outcome for socket preservation. This calculation was originally based on a comparison of two independent means using data reported by Osman et al. [28], who reported mean ridge widths of 5.75 ± 0.95 mm and 6.95 ± 1.17 mm after 3 months. With $\alpha = 0.05$ and 80% power, the minimum required sample size was calculated as 13 participants per group. To compensate for an anticipated dropout rate of approximately 20%, 18 participants were allocated to each group, resulting in a total sample size of 72 participants.

Since the final study design included four groups and repeated measurements, a post hoc power analysis was

also performed using G*Power to evaluate the adequacy of the achieved sample size for the four-arm repeated-measures design. Using α = 0.05, four groups, three measurement time points, and an observed effect size based on partial eta squared of 0.686, the achieved statistical power was 0.99. Therefore, although the original sample size estimation was based on a two-group comparison, the achieved sample size was considered sufficient for detecting the observed differences across the four groups.

$$n = \frac{\left(S_1^2 + S_2^2\right) * \left(Z_{1-\frac{\alpha}{2}} + Z_{1-\beta}\right)^2}{\left(\mu_1 - \mu_2\right)^2} = \frac{\left(0.95^2 + 1.17^2\right) * (2 + 0.84)^2}{(5.75 - 6.95)^2} \cong 13$$

Randomization and blinding

The patients were randomly assigned to four groups (T1, T2, T3, and C) by block randomization. The process was done using following website: https://www.sealedenvelope.com/randomisation/simulation. Finally, 18 partic ipants were allocated to each group, resulting in a total sample size of 72 participants. In this single-blind clinical trial, the outcome investigator (A.M., periodontist) was blinded to group allocation throughout the study. The administering clinician (Am.Mo, oral and maxillofacial surgeon) performed all interventions (atraumatic extractions, PRF preparation/application, and PBM irradiations) and was aware of group assignments, as was necessary for treatment delivery. Patients were not blinded due to visible differences in interventions. To maintain blinding, group allocation was concealed from the outcome assessor via sealed envelopes and separate scheduling for follow-up assessments.

Examiner calibration

Prior to the start of data collection, the periodontist (A.M.) and oral radiologist (M.A.) underwent calibration training for all clinical and radiographic measurements to ensure standardized assessment. This included repeated measurements on a pilot set of 10 cases which were not included in the main study to achieve consistency. Inter-examiner agreement was assessed via intraclass correlation coefficient (ICC) on these pilot cases, yielding values > 0.90 for all parameters, indicating excellent reliability.

For intra-examiner reliability verification during the study, a random subset of measurements (20% of cases for each parameter) was re-evaluated by the same examiners on two separate occasions, separated by at least one week, in a blinded manner. The ICC was calculated for each variable using a two-way mixed-effects model for absolute agreement. ICC values were 1.00 for buccolingual ridge width, ridge height, and soft tissue healing, and 0.95 (range 0.81–0.99) for keratinized gingiva width (KGW) ($P < 0.001$ for all). Since all ICC values exceeded 0.95, the measurements demonstrated excellent intra-examiner reliability throughout the study.

Intervention

An oral and maxillofacial surgeon employed a Periotome (SEDRADENT, Cairo, Egypt) and instruments (DeNA Life Science Inc., Tokyo, Japan) to perform tooth extraction atraumatically after soliciting written informed consent from the patients. The extraction cavity was subsequently curetted and cleansed with saline. Participants were randomly assigned to one of four groups, with 18 participants in each group: PBM, PRF, PRF-PBM, or control.

Control group

A #19 needle and 4 – 0 polyglycolic acid sutures were used to stitch the extraction socket (Sutures, Iran). In this group, no intervention was carried out.

PBM group

After proper suturing of the sockets, for enhancement of soft tissue healing, a diode laser (Sirona Dentsply, Germany) with 660 nm wavelength, 100 mW power, and 2 J/cm² energy density was irradiated for 10 s in continuous-wave mode. The laser was irradiated from the buccal and lingual aspects obliquely toward the tissue margins and also at the closest possible distance from the socket. Next, for enhancement of hard tissue healing, a diode laser with 970 nm wavelength, 200 mW power, and 6 J/cm² energy density was irradiated for 20 s using a low-level laser tip once from the buccal and once from the lingual aspect, such that half of the root length was covered in each cycle. This laser irradiation protocol was performed 1 day after extraction and then every other day for a minimum of 2 weeks (days 1, 3, 6, 8, 10, 13, and 15) [11, 29–31] (Fig. 2).

PRF group

A nurse collected 10 mL of intravenous blood from the median cubital vein of each patient and transferred it to a dry test vial that was devoid of anticoagulant. Subsequently, it was centrifuged at 3000 rpm for 10 min (Atarin Teb Vasna, Iran) [32, 33]. The PRF was separated using a sterile tweezer, applied to the cavity, and sutured using a figure-of-eight suture with a #19 needle and 4 – 0 polyglycolic acid suture thread (Sutures, Iran).

PRF-PBM group

In this group, PRF was applied to the extraction socket in the same way as in the PRF group, followed by PBM irradiation as described for the PBM group.

The patients subsequently received postoperative instructions and were prescribed 400 mg ibuprofen twice

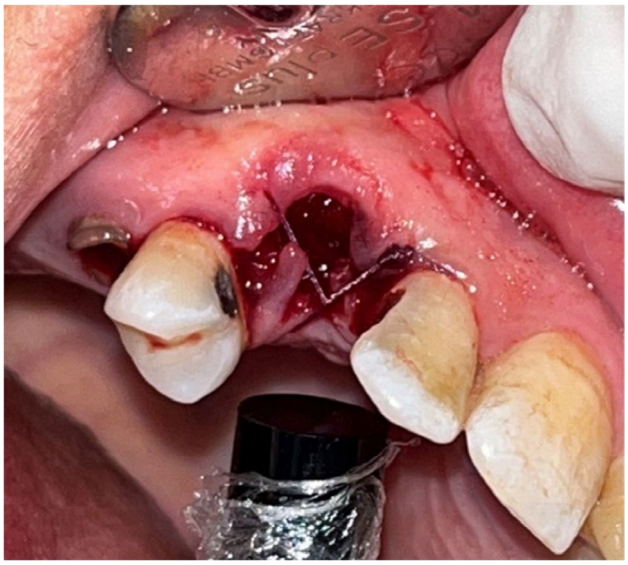


Fig. 2 Irradiation of diode laser with the closest distance from the socket for enhancement of tissue healing

a day for 3 days, 500 mg amoxicillin three times a day for 5 days, and 0.2% chlorhexidine rinse for 2 weeks [34].

Radiographic and clinical assessment

Assessment of radiographic parameters

Preoperative panoramic radiographs, together with clinical examination, were used for initial eligibility assessment. For radiographic outcome assessment, CBCT scans were obtained immediately after atraumatic extraction and again 8 weeks after extraction using a Carestream Dental CBCT unit (Carestream Dental LLC, Atlanta, GA, USA; 6.3 mA, 90 kVp, 5 × 5 cm field of view, 90 µm voxel size, and 20 s exposure time). The immediate post-extraction CBCT scan served as the baseline radiographic measurement, and the 8-week CBCT scan was used for follow-up comparison. The vertical height and buccolingual width of alveolar ridge were assessed on cross-sectional slices with 0.5 mm slice thickness and 2 mm slice intervals. CS Imaging software was used for imaging, the files were saved in DICOM format, and the measurements were made with OnDemand software. CBCT scans were obtained and interpreted by an oral radiologist. The assessed parameters were as follows:

(I) Vertical bone height in the anterior maxilla: A line was drawn parallel to the longitudinal axis of the alveolar ridge from the alveolar crest to the nasal cavity floor [35].
(II) Vertical bone height in the anterior mandible: From the alveolar bone crest to the endpoint of the cancellous bone [23].
(III) Bone width: From the buccal cortical plate to the palatal/lingual cortical plate [36].

Buccolingual ridge width was measured on cross-sectional CBCT images at the mesiodistal midpoint of the extraction socket. The midpoint was first identified on the axial view as the center of the extraction socket in the mesiodistal dimension, and the corresponding cross-sectional slice was selected for measurement. On this slice, a vertical reference line was drawn parallel to the long axis of the socket/alveolar ridge. The alveolar crest was defined as the most coronal point of the remaining buccal and palatal/lingual bony walls. To standardize the apico-coronal level, buccolingual ridge width was measured at 3 mm apical to the alveolar crest, perpendicular to the long axis of the socket, from the outer surface of the buccal cortical plate to the outer surface of the palatal/lingual cortical plate. All radiographic parameters were measured by an oral radiologist twice: immediately after extraction and 8 weeks after extraction (Fig. 3).

Assessment of clinical parameters

The following clinical parameters were also evaluated by a periodontist:

(I) Keratinized gingiva width (KGW) from the mid-facial point of gingival margin to the mucogingival junction of respective tooth before extraction

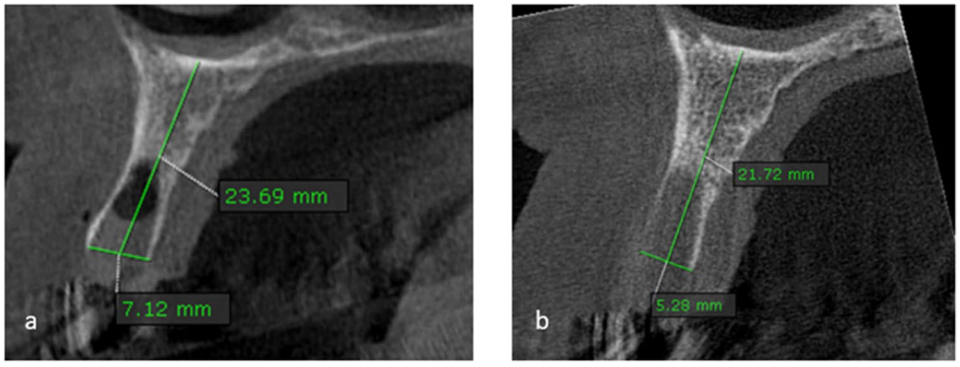


Fig. 3 CBCT images taken **a** immediately and **b** 8 weeks after extraction of a maxillary tooth

(baseline), and from the top of the bone crest (after tooth extraction) to mucogingival junction 8 weeks after the extraction was measured using an UNC-15 periodontal probe (Hu-Friedy, Chicago, IL, USA) [1].

(II) The quality of soft tissue healing was clinically assessed at 3, 7, and 14 days after tooth extraction. For this purpose, modified healing index was used, which has 3 scores for 4 parameters of bleeding, pus discharge, tissue color, and healing tissue consistency. A score of 4 indicated excellent healing while a score of 12 indicated severe impairment in healing [37].

(III) The pain level was quantified using a 10-point visual analog scale (VAS) at 24, 48, and 72 h after extraction [37]. Score 0 indicated no pain or discomfort, scores 1–4 indicated mild pain, scores 5–6 indicated moderate pain, and scores 7–10 indicated severe pain.

Statistical analysis

The Shapiro-Wilk test was used to assess the normality of continuous variables. For normally distributed repeated continuous outcomes, changes over time and differences among groups were analyzed using repeated-measures ANOVA, followed by Bonferroni-adjusted post hoc comparisons when appropriate. Because the soft tissue healing score did not meet the assumption of normality and showed a ceiling effect, particularly at later follow-up points, this outcome was re-analyzed using a generalized estimating equation (GEE) model. For buccolingual ridge width, an additional ANCOVA model was performed to compare follow-up ridge width among the four groups. Statistical analyses were performed using SPSS version 22. The significance level was set at 0.05.

Results

A total of 72 patients were assessed, with 56.9% males ($n = 41$) and 43.1% females ($n = 31$). The results of the Chi-square test indicated a statistically significant difference in the gender distribution of the groups. The patients had a mean age of 46.58 ± 8.88 years, ranging from 26 to 65 years old. The results of the one-way ANOVA indicated that the difference in mean age between the four groups was not statistically significant (Table 1).

Because gender distribution differed significantly among the groups, a sensitivity analysis was performed by repeating the primary analyses with gender entered as an additional covariate. The inclusion of gender did not materially change the direction or statistical significance of the main outcomes, and gender was not a significant predictor of the primary clinical or radiographic outcomes. Therefore, the observed gender imbalance did not appear to substantially affect the main conclusions of the study.

Primary outcomes

Pain

Table 2 illustrates that the mean VAS pain score in all groups experienced a substantial change over time, as evidenced by the within-group assessments ($P < 0.001$). Regardless of the intervention type, VAS pain scores decreased over time in all four groups. The mean VAS pain score of the groups differed significantly ($P < 0.001$), as indicated by the between-group assessments. There was no significant difference in pain scores between the PBM and PRF groups after 24–48 h ($P > 0.05$). The experimental groups had considerably lower mean pain scores than the control group at 24, 48, and 72 h ($P < 0.05$). The PRF-PBM group had lower pain scores than the PBM and PRF groups at all time periods, however only the PBM group showed a statistically significant difference ($P < 0.05$) (Fig. 4).

KGW

The mean KGW decreased substantially over time in all four groups (Table 2), suggesting that time has a substantial impact on KGW, regardless of the type of intervention ($P < 0.05$). The mean KGW did not differ substantially among the four groups, suggesting that the treatment type had an insignificant impact on KGW ($P > 0.05$).

Table 1 The demographic characteristics of the participants

			GROUP				Total	p-value
			PRF	PBM	PRF + PBM	Control		
SEX	M	Count	8	15	11	7	41	0.032*
		% within GROUP	44.4%	83.3%	61.1%	38.9%	56.9%	
	F	Count	10	3	7	11	31	
		% within GROUP	55.6%	16.7%	38.9%	61.1%	43.1%	
Age		Mean	43.66	50.88	45.77	46.00	46.58	0.091**
		Std. Deviation	8.79	7.68	9.14	8.91	8.87	

* Chi-square

** one-way ANOVA

Table 2 Mean VAS pain score, KGW, and alveolar bone height and width in the four groups at different time points

Variable		Mean ± SD					Partial Eta Squared	*P*-value	
		Total	PRF	PBM	PRF + PBM	Control		Time effect	Treatment effect
VAS	24 h	2.14 ± 1.64	1.61 ± 1.75	2.44 ± 0.92	1.17 ± 0.71	3.33 ± 1.97	0.686	< 0.001	< 0.001
	48 h	0.94 ± 1.18	0.78 ± 1.21	1.00 ± 0.59	0.11 ± 0.47	1.89 ± 1.45			
	72 h	0.18 ± 0.42	0.17 ± 0.38	0.11 ± 0.32	0	0.44 ± 0.62			
KGW	before	4.86 ± 1.35	5.44 ± 1.38	4.5 ± 1.58	4.89 ± 1.18	4.83 ± 1.54	0.784	< 0.001	0.317
	after	3.39 ± 1.34	3.83 ± 1.25	3.39 ± 1.19	3.72 ± 1.27	3.06 ± 1.35			
Height (mm)	before	18.87 + 5.00	17.65 + 4.95	19.06 + 4.25	19.46 ± 3.85	19.31 ± 6.69	0.764	< 0.001	0.684
	after	17.72 + 5.15	16.64 + 4.88	18.26 + 4.37	18.83 ± 3.81	17.16 ± 7.07			
Width (mm)	before	7.24 ± 1.37	7.04 ± 1.09	7.06 ± 1.40	7.76 ± 1.60	7.13 ± 1.34	0.875	< 0.001	0.038
	after	6.09 ± 1.51	5.92 ± 1.4	6.27 ± 1.41	7.10 ± 1.72	5.17 ± 1.11			

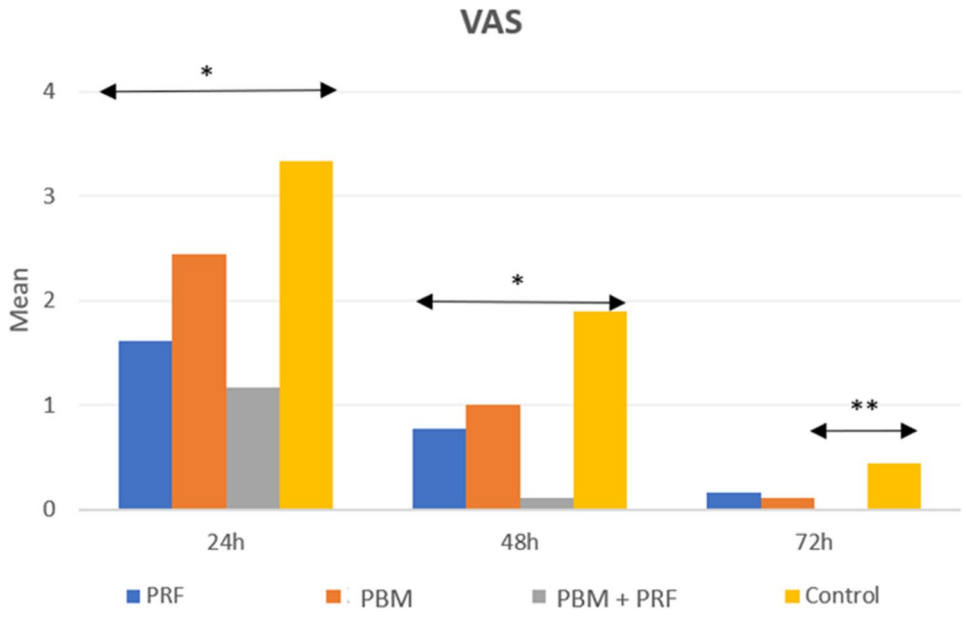


Fig. 4 Pairwise comparisons of the groups regarding the pain score using the Bonferroni post-hoc test (*: *P*-value < 0.05, **: *P*-value < 0.001)

Alveolar ridge height

In all four groups, the mean alveolar ridge height decreased substantially after 8 weeks, as demonstrated in Table 2. This finding suggests that time has a significant impact on this parameter, regardless of the intervention type ($P < 0.05$). Nevertheless, the four groups did not exhibit a significant difference in alveolar ridge height compared to one another, suggesting that the treatment type had an insignificant impact on this parameter ($P > 0.05$).

Alveolar ridge buccolingual width

The mean buccolingual ridge width decreased from baseline to 8 weeks in all groups, indicating post-extraction horizontal ridge remodeling. Because baseline ridge width may influence follow-up ridge dimensions, an ANCOVA model was performed with follow-up ridge width as the dependent variable and baseline ridge width as a covariate. Baseline ridge width was strongly associated with follow-up ridge width (B = 0.946, 95% CI: 0.868 to 1.025, $P < 0.001$), indicating that initial ridge dimensions significantly influenced the 8-week measurements. After adjustment for baseline ridge width, all intervention groups showed significantly greater follow-up ridge width compared with the control group: PRF (B = 0.730, 95% CI: 0.435 to 1.024, $P < 0.001$), PBM (B = 1.164, 95% CI: 0.869 to 1.458, $P < 0.001$), and PRF + PBM (B = 1.328, 95% CI: 1.030 to 1.627, $P < 0.001$). The largest adjusted effect was observed in the PRF + PBM group. These findings indicate that the interventions were associated with better preservation of buccolingual ridge width after adjustment for baseline ridge width (Table 3).

Soft tissue healing

Since the soft tissue healing score was not normally distributed and showed a ceiling effect at later follow-up points, this variable was analyzed using a GEE model (Table 4). In this model, the control group and the 72-hour assessment were used as reference categories. The results showed that healing scores were significantly

Table 3 ANCOVA parameter estimates for follow-up buccolingual ridge width adjusted for baseline ridge width

Parameter*	B	Std. Error	t	*P*-value	95% CI Lower	95% CI Upper	Partial Eta Squared
Intercept	−1.573	0.298	−5.278	< 0.001	−2.168	−0.978	0.294
PRF vs. control	0.730	0.147	4.948	< 0.001	0.435	1.024	0.268
PBM vs. control	1.164	0.147	7.890	< 0.001	0.869	1.458	0.482
PRF + PBM vs. control	1.328	0.150	8.884	< 0.001	1.030	1.627	0.541
Baseline ridge width	0.946	0.039	24.151	< 0.001	0.868	1.025	0.897

* The dependent variable was follow-up buccolingual ridge width at 8 weeks. Baseline ridge width was entered as a covariate. The control group was used as the reference category

Table 4 Generalized estimating equation model for soft tissue healing scores

Parameter*	B	Std. Error	95% Wald CI Lower	95% Wald CI Upper	Wald Chi-square	df	*P*-value
Intercept	1.510	0.0212	1.468	1.551	5071.789	1	< 0.001
PRF vs. control	−0.025	0.0208	−0.066	0.016	1.466	1	0.226
PBM vs. control	−0.051	0.0205	−0.091	−0.011	6.135	1	0.013
PRF + PBM vs. control	−0.064	0.0161	−0.096	−0.032	15.776	1	< 0.001
Day 14 vs. 72 h	−0.088	0.0157	−0.119	−0.057	31.440	1	< 0.001
Day 7 vs. 72 h	−0.074	0.0123	−0.099	−0.050	36.678	1	< 0.001
Scale	0.007	—	—	—	—	—	—

* The dependent variable was soft tissue healing score. Lower scores indicate better healing. The model included group and time as predictors and accounted for repeated measurements within participants. The control group and the 72-hour time point were used as reference categories

lower in the PBM group compared with the control group (B = – 0.051, 95% CI: –0.091 to – 0.011, $P = 0.013$) and in the PRF-PBM group compared with the control group (B = – 0.064, 95% CI: –0.096 to – 0.032, $P < 0.001$). The difference between the PRF group and the control group was not statistically significant (B = – 0.025, 95% CI: –0.066 to 0.016, $P = 0.226$). Compared with the 72-hour reference point, healing scores were significantly lower at day 7 (B = – 0.074, 95% CI: –0.099 to – 0.050, $P < 0.001$) and day 14 (B = – 0.088, 95% CI: –0.119 to – 0.057, $P < 0.001$).

Discussion

The present randomized clinical trial compared PBM, PRF, and their combined application for socket preservation after atraumatic extraction of anterior and premolar teeth. The findings suggest that the intervention groups provided clinical and radiographic benefits compared with the control condition, although the pattern of benefit differed by outcome. PRF-PBM was associated with the lowest postoperative pain scores and the largest adjusted effect for buccolingual ridge width, while PBM and PRF-PBM showed significantly better soft tissue healing scores than the control group in the GEE analysis. No significant intergroup differences were observed for KGW or vertical ridge height after 8 weeks.

The PRF-PBM group showed the lowest postoperative pain scores at all time intervals, although the difference was statistically significant only compared with the PBM group. Similarly, Peimani et al. [38] reported that PBM (808 nm, 100 nW) enhanced pain alleviation after tooth extraction in rodents. Moreover, Alasgah et al. [39] indicated the potential for PRF and PRF in combination with collagen to reduce VAS following tooth extraction. This finding may be explained by the complementary biological effects of PRF and PBM. PBM may reduce pain through modulation of inflammation, improved microcirculation, increased mitochondrial ATP production, and effects on neural mediators such as beta-endorphin and serotonin [40–43]. PRF provides an autologous fibrin matrix containing platelets, leukocytes, cytokines, and growth factors, which may stabilize the clot, regulate inflammation, and support early wound healing [5, 44, 45]. Therefore, PRF may provide a biologically active scaffold, while PBM may stimulate cellular activity and reduce early inflammatory pain responses.

In contrast, Demirok et al. [46] did not observe any difference in pain levels of PRF and PBM after third molar extraction. The absence of significant differences between PRF-PBM and PRF alone for some pain comparisons may be related to the generally low VAS scores, the rapid spontaneous reduction of pain after atraumatic extraction, and the limited sample size.

In the GEE analysis, PBM and PRF-PBM were associated with significantly better soft tissue healing scores compared with the control group, whereas PRF alone did not differ significantly from the control group. These findings support a possible role of PBM, alone or combined with PRF, in enhancing early soft tissue healing after extraction. PBM can enhance fibroblast activity, collagen synthesis, angiogenesis, and epithelial cell migration, which are particularly important in the first few days after extraction [47, 48]. Similar to the present findings, Srinivas et al. [49] reported a higher healing index in the PRF group compared to the no-PRF control group after tooth extraction. PRF may contribute through gradual

release of growth factors, including platelet-derived growth factor, transforming growth factor-beta, and vascular endothelial growth factor, which are involved in angiogenesis and tissue regeneration [9, 13, 44, 50]. However, PRF alone did not show a significant improvement compared with the control group. This may be due to the timing of healing assessment, variability in PRF quality and quantity, and spontaneous healing in sockets with intact walls.

The interpretation of soft tissue healing should also consider the ceiling effect observed in this variable. By day 14, healing scores approached the best possible value in all groups, which limited the ability to detect additional between-group differences at later time points. Therefore, the GEE analysis was used to account for repeated observations and the non-normal distribution of healing scores. This analysis confirmed significantly better healing scores in the PBM and PRF-PBM groups compared with the control group, whereas PRF alone did not differ significantly from the control group.

The results demonstrated a substantial decrease in ridge width across all groups at 8 weeks, indicating that none of the interventions completely prevented post-extraction horizontal remodeling. However, after adjustment for baseline ridge width using ANCOVA, all intervention groups showed significantly greater follow-up buccolingual ridge width compared with the control group, with the largest adjusted effect observed in the PRF-PBM group. Horizontal ridge resorption is a major concern after extraction, especially when implant placement is planned. It has been suggested that narrow buccal plates are more susceptible to post-extraction resorption [51]. The improved adjusted ridge-width outcomes in the intervention groups may be related to enhanced early wound stability and biological activity within the socket. PRF may stabilize the blood clot and act as a temporary fibrin scaffold for migration of fibroblasts, endothelial cells, and osteogenic cells. PBM may enhance mitochondrial respiration, ATP production, osteoblast proliferation and differentiation, collagen formation, and angiogenesis [10, 51, 52]. The larger adjusted effect in the PRF-PBM group may reflect the complementary action of PRF as a scaffold and growth factor reservoir together with the biostimulatory effects of PBM. Nevertheless, these interventions should be interpreted as reducing horizontal dimensional loss rather than completely preventing physiologic bone remodeling.

The present findings are partly consistent with previous studies reporting beneficial effects of PBM or PRF-based approaches on socket healing and ridge preservation. However, direct comparison is difficult because of substantial methodological heterogeneity. Osman et al. [28] reported favorable results for laser-assisted ridge preservation combined with PRF and/or grafting materials, but the use of bone grafts provides an osteoconductive and space-maintaining scaffold that was not used in the present study. Abolfazli et al. [1] reported lower horizontal ridge resorption with PBM, whereas Zhang et al. [27] and Suttapreyasri and Leepong [53] found limited dimensional benefits for PRF alone. These discrepancies may be explained by differences in laser wavelength, energy density, irradiation schedule, PRF preparation protocol, socket anatomy, use or absence of grafting materials, measurement methods, and follow-up duration. Thus, inconsistent findings across studies may reflect differences in protocols rather than true contradiction.

In the present study, no significant intergroup differences were found for vertical ridge height. This finding is consistent with studies reporting limited vertical preservation after PRF or PBM alone [1, 23, 39], but differs from studies in which graft materials were combined with PRF and/or laser therapy [28]. This may be because vertical remodeling after extraction is influenced by socket anatomy, buccal plate thickness, flap design, and the natural resorption of bundle bone. Although PRF and PBM may enhance early biological healing, they do not provide the mechanical space-maintaining effect of bone grafts or barrier membranes. Therefore, in the absence of grafting materials, these interventions may be insufficient to prevent vertical dimensional loss. This may explain why ridge height decreased significantly over time in all groups, while no treatment effect was detected. The evaluation of KGW showed a significant reduction over time in all groups; however, no significant difference was detected among the four groups at 8 weeks. This finding suggests that PBM, PRF, or their combination did not significantly influence keratinized gingival width within the short follow-up period of the present study. This may be explained by the fact that KGW is mainly affected by gingival phenotype, mucogingival anatomy, wound contraction, flap design, and the position of the mucogingival junction, rather than by early socket healing alone. Although PBM and PRF may accelerate soft tissue repair, they are not primarily mucogingival augmentation procedures and may not be sufficient to preserve or increase KGW, especially when no soft tissue grafting procedure is performed. In contrast, Abolfazli et al. [1] reported higher KGW in the PBM group than in the control group, and Thalaimalai et al. [54] found greater improvement in periodontal parameters in the PRF + laser group than in the PRF group after 6 months. These discrepancies may be related to differences in study design, baseline gingival phenotype, surgical technique, PRF volume and preparation protocol, laser wavelength and energy density, and particularly the duration of follow-up.

Overall, the findings suggest that PRF-PBM may provide an early biological advantage for socket healing, especially in terms of postoperative comfort, early soft

tissue healing, and horizontal ridge preservation. The combined protocol may work by integrating the scaffold and growth factor reservoir of PRF with the cellular activation, angiogenic stimulation, and anti-inflammatory effects of PBM. However, this effect should be interpreted as an early healing benefit rather than complete ridge preservation. The 8-week follow-up period was sufficient to assess early pain, soft tissue healing, and initial radiographic changes, but it may not fully evaluate later alveolar bone remodeling. Another limitation is that the original sample size calculation was based on a two-group comparison rather than the final four-arm repeated-measures design. Although a post hoc power analysis suggested adequate achieved power, this retrospective assessment should be interpreted cautiously, and future studies should prospectively calculate sample size based on the intended four-arm design and primary outcome. Another limitation of the present study was the imbalance in gender distribution among the four groups. Although the sensitivity analysis adjusted for gender showed that this imbalance did not materially affect the main outcomes or alter the primary conclusions, the unequal distribution of male and female participants should still be considered when interpreting the findings. Larger randomized trials with longer follow-up, balanced baseline characteristics, and prospectively defined statistical analyses are recommended to confirm these results.

Conclusion

Within the limitations of this randomized clinical trial, PBM, PRF, and PRF-PBM were associated with favorable early outcomes after atraumatic tooth extraction compared with the control condition, although the effects differed across outcomes. PRF-PBM showed the lowest postoperative pain scores and the largest adjusted effect for buccolingual ridge width preservation. In the GEE analysis, PBM and PRF-PBM were associated with significantly better early soft tissue healing scores than the control group, whereas PRF alone did not show a significant healing advantage. After adjustment for baseline ridge width, all intervention groups showed significantly greater follow-up buccolingual ridge width than the control group. No significant intergroup differences were observed for KGW or vertical ridge height at 8 weeks. These findings suggest that PRF-PBM may provide an early biological advantage for socket healing and horizontal ridge preservation, but longer-term studies are required to determine whether these benefits are maintained during later bone remodeling and whether they improve subsequent implant-related outcomes.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions
Av.Ma. and Am. Mo. Conceptualized the study. R.F. and M.S. supervised and validated the study. M.M. and Av.Ma. prepared the manuscript. F.S. And Z.K. collected data. N.H. and M.A. Visualization Data. R.N. analyzed data. All authors reviewed the manuscript.

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Data availability
The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted at the Periodontology Department of the School of Dentistry, Mazandaran University of Medical Sciences. This study adhered to the Declaration of Helsinki. The study protocol received ethical approval from the Ethics Committee of Mazandaran University of Medical Sciences (IR.MAZUMS.REC.1401.497 and IR.MAZUMS.REC.1401.427) and the Iranian Registry of Clinical Trials (IRCT20230305057619N registered on 2023-05-13, and IRCT20230222057497N1 registered on 2023-05-09). An informed written consent was taken from the patients. https://irct.behdasht.gov.ir/trial/69294. https://irct.behdasht.gov.ir/trial/69308.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Abolfaazli N, Kavoosi F, Saleh Saber F, Sedighi Shamami M, Abedini M. Clinical and histological evaluation of low level laser therapy on ridge preservation after tooth extraction. Laser Med. 2016;13(1):15–7.

2. Amaroli A, Colombo E, Zekiy A, Aicardi S, Benedicenti S, De Angelis N. Interaction between laser light and osteoblasts: photobiomodulation as a trend in the management of socket bone preservation—a review. Biology. 2020;9(11):409.
3. Anwandter A, Bohmann S, Nally M, Castro AB, Quirynen M, Pinto N. Dimensional changes of the post extraction alveolar ridge, preserved with Leukocyte-and Platelet Rich Fibrin: A clinical pilot study. J Dent. 2016;52:23–9.
4. Corning PJ, Mealey BL. Ridge preservation following tooth extraction using mineralized freeze-dried bone allograft compared to mineralized solvent-dehydrated bone allograft: A randomized controlled clinical trial. J Periodontol. 2019;90(2):126–33.
5. Pan J, Xu Q, Hou J, Wu Y, Liu Y, Li R, et al. Effect of platelet-rich fibrin on alveolar ridge preservation: a systematic review. J Am Dent Association. 2019;150(9):766–78.
6. Serrano Méndez CA, Lang NP, Caneva M, Ramírez Lemus G, Mora Solano G, Botticelli D. Comparison of allografts and xenografts used for alveolar ridge preservation. A clinical and histomorphometric RCT in humans. Clin Implant Dent Relat Res. 2017;19(4):608–15.
7. Daigo Y, Daigo E, Fukuoka H, Fukuoka N, Ishikawa M, Takahashi K. Wound healing and cell dynamics including mesenchymal and dental pulp stem cells induced by photobiomodulation therapy: an example of socket-preserving effects after tooth extraction in rats and a literature review. Int J Mol Sci. 2020;21(18):6850.
8. El-Hayes KA, Zaky AA, Ibrahim ZA, Allam GFA, Allam MF. Usage of low level laser biostimulation and platelet rich fibrin in bone healing: Experimental study. Dent Med Probl. 2016;53(3):338–44.
9. Reis CHB, Buchaim DV, Ortiz AC, Fideles SOM, Dias JA, Miglino MA, et al. Application of fibrin associated with photobiomodulation as a promising strategy to improve regeneration in tissue engineering: a systematic review. Polymers. 2022;14(15):3150.
10. Jamalpour MR, Shahabi S, Baghestani M, Shokri A, Jamshidi S, Khazaei S. Complementarity of surgical therapy, photobiomodulation, A-PRF and L-PRF for management of medication-related osteonecrosis of the jaw (MRONJ): an animal study. BMC Oral Health. 2022;22(1):241.
11. Križaj Dumić A, Pajk F, Olivi G. The effect of post-extraction socket preservation laser treatment on bone density 4 months after extraction: Randomized controlled trial. Clin Implant Dent Relat Res. 2021;23(3):309–16.
12. Azangookhiavi H, Ghodsi S, Jalil F, Dadpour Y. Comparison of the efficacy of platelet-rich fibrin and bone allograft for alveolar ridge preservation after tooth extraction: a clinical trial. Front Dentistry. 2020;17:1.
13. Serafini G, Lollobrigida M, Fortunato L, Mazzucchi G, Lamazza L, Di Nardo D, et al. Postextractive alveolar ridge preservation using L-PRF: clinical and histological evaluation. Case Rep dentistry. 2020;2020(1):5073519.
14. Moaddabi A, Spagnuolo G, Karimzadegan A, Papi S, Hosseinnataj A, Maboudi A,Semnani A, Soltani P, Rengo C. The Effect of Platelet-Rich Fibrin in Preventing Inferior Alveolar Nerve Paresthesia After Mandibular Third Molar Surgery: A Randomized Clinical Trial. European Journal of Dentistry. 2026;17.
15. Moaddabi A, Soltani P, Yazdani A, Nikbakht MH, Amani Beni P,Modabber E, Iaculli F, Spagnuolo G. Application of Platelet-Rich Fibrin and Bone Morphogenetic Protein for Full-Mouth Implant-Based OralRehabilitation in a Case of Mandibular Osteoradionecrosis. Case reports in dentistry. 2023;2023(1):2449298.
16. Alrayyes Y, Al-Jasser R. Regenerative potential of platelet Rich Fibrin (PRF) in Socket Preservation in comparison with Conventional Treatment modalities: a systematic review and Meta-analysis. Tissue Eng Regenerative Med. 2022;19(3):463–75.
17. Fekrazad S, Fekrazad R. Photobiomodulation: the forth side of tissue engineering foursquare. J Lasers Med Sci. 2022;13:e73.
18. Hopewell S, Chan A-W, Collins GS, Hróbjartsson A, Moher D, Schulz KF, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. Lancet. 2025;405(10489):1633–40.
19. Mendoza-Azpur G, Olaechea A, Padial-Molina M, Gutiérrez-Garrido L, O'Valle F, Mesa F, et al. Composite alloplastic biomaterial vs. autologous platelet-rich fibrin in ridge preservation. J Clin Med. 2019;8(2):223.
20. Girish Kumar N, Chaudhary R, Kumar I, Arora SS, Kumar N, Singh H. To assess the efficacy of socket plug technique using platelet rich fibrin with or without the use of bone substitute in alveolar ridge preservation: a prospective randomised controlled study. Oral maxillofacial Surg. 2018;22:135–42.
21. Areewong K, Chantaramungkorn M, Khongkhunthian P. Platelet-rich fibrin to preserve alveolar bone sockets following tooth extraction: a randomized controlled trial. Clin Implant Dent Relat Res. 2019;21(6):1156–63.
22. Dhamija R, Shetty V, Vineeth K, Nagaraju R, Rao RS. Socket preservation with demineralized freeze-dried bone allograft and platelet-rich fibrin for implant site development: A randomized controlled trial. J Indian Prosthodontic Soc. 2020;20(3):304–11.
23. Abdel Mageed H, Hanna R, Benedicenti S. The effectiveness of 980 nm laser therapy on socket preservation: a preliminary split mouth randomized controlled clinical study. J Dent Oral Disord. 2019;5(2):1116.
24. De Angelis P, De Angelis S, Passarelli PC, Liguori MG, Manicone PF, D'Addona A. Hard and soft tissue evaluation of different socket preservation procedures using leukocyte and platelet-rich fibrin: a retrospective clinical and volumetric analysis. J Oral Maxillofac Surg. 2019;77(9):1807–15.
25. Dhamija R, Shetty V, Vineeth K, Nagaraju R, Rao RS. Socket preservation with demineralized freeze-dried bone allograft and platelet-rich fibrin for implant site development: A randomized controlled trial. J Indian Prosthodontic Soc. 2020;20(3):304.
26. Clark D, Rajendran Y, Paydar S, Ho S, Cox D, Ryder M, et al. Advanced platelet-rich fibrin and freeze-dried bone allograft for ridge preservation: a randomized controlled clinical trial. J Periodontol. 2018;89(4):379–87.
27. Zhang Y, Ruan Z, Shen M, Tan L, Huang W, Wang L, et al. Clinical effect of platelet-rich fibrin on the preservation of the alveolar ridge following tooth extraction. Experimental therapeutic Med. 2018;15(3):2277–86.
28. Osman RA, Elkilani NS, Adawy HA. Evaluation of laser assisted ridge preservation combined with platelet rich fibrin and/or bone. Al-Azhar Dent J Girls. 2022;9(1):135.
29. Abd-Elhaleem Othman MA, Zaky AA, Eltayeb EA, Khalil NM. A radiographic and histological study to compare red (650 nm) versus near infrared (810 nm) diode lasers photobiomodulation for alveolar socket preservation. Sci Rep. 2024;14(1):6871.
30. Hazzaa HH, Shiekh MAE, AbdAllah MF, Abdelgawad N, El-Mahdy MA, Shoshan HS et al. Laser-assisted socket seal surgery using bioactive glass for dental implant site development: a randomized clinical trial. Sci Rep. 2025;15(1):43386. https://doi.org/10.1038/s41598-025-28824-7.
31. Niedzielska I, Dawiec G, Wiench R, Pihut M, Skaba D, Arnabat-Dominguez J. Laser-based photobiomodulation in postoperative tissue healing in oral and maxillofacial surgery: systematic review of RCTs. J Clin Med. 2026;15(2):613.
32. Al-Rihaymee S, Mahmood MS. Platelet-rich fibrin potential role in periodontal regeneration: A review study. Med J Babylon. 2023;20(1):1–6.
33. Das S, Jhingran R, Bains VK, Madan R, Srivastava R, Rizvi I. Socket preservation by beta-tri-calcium phosphate with collagen compared to platelet-rich fibrin: A clinico-radiographic study. Eur J dentistry. 2016;10(02):264–76.
34. Yewale M, Bhat S, Kamath A, Tamrakar A, Patil V, Algal AS. Advanced platelet-rich fibrin plus and osseous bone graft for socket preservation and ridge augmentation–A randomized control clinical trial. J Oral Biology Craniofac Res. 2021;11(2):225–33.
35. Zhang W, Skrypczak A, Weltman R. Anterior maxilla alveolar ridge dimension and morphology measurement by cone beam computerized tomography (CBCT) for immediate implant treatment planning. BMC Oral Health. 2015;15:1–8.
36. Um IW, Kim YK, Park JC, Lee JH. Clinical application of autogenous demineralized dentin matrix loaded with recombinant human bone morphogenetic-2 for socket preservation: A case series. Clin Implant Dent Relat Res. 2019;21(1):4–10.
37. Marenzi G, Riccitiello F, Tia M, di Lauro A, Sammartino G. Influence of leukocyte-and platelet-rich fibrin (L-PRF) in the healing of simple postextraction sockets: a split-mouth study. BioMed research international. 2015(1):369273. https://doi.org/10.1155/2015/369273.
38. Peimani A, Abedi P, Hajjafari H. Effects of low-level laser therapy on tooth socket repair in diabetic male rats: an animal study. J Res Dent Maxillofacial Sci. 2020;5(3):37–42.
39. Alasqah M, Alansary RD, Gufran K. Efficacy of platelet-rich fibrin in preserving alveolar ridge volume and reducing postoperative pain in site preservation of post-extracted sockets. Medicina. 2024;60(7):1067.
40. de Sousa MVP, Kawakubo M, Ferraresi C, Kaippert B, Yoshimura EM, Hamblin MR. Pain management using photobiomodulation: mechanisms, location, and repeatability quantified by pain threshold and neural biomarkers in mice. J Biophotonics. 2018;11(7):e201700370.
41. dos Santos Santinoni C, Oliveira HFF, de Souza Batista VE, Lemos CAA, Verri FR. Influence of low-level laser therapy on the healing of human bone maxillofacial defects: A systematic review. J Photochem Photobiol B. 2017;169:83–9.
42. Kotb S. Photobiomodulation and its application in dentistry. Advances in Medical. Dent Health Sci. 2022;5(3). 29-33. https://doi.org/10.31219/osf.io/45ncs.

43. Wang CY, Tsai SC, Yu MC, Lin YF, Chen CC, Chang PC. Light-emitting diode irradiation promotes donor site wound healing of the free gingival graft. J Periodontol. 2015;86(5):674–81.
44. Al-Maawi S, Becker K, Schwarz F, Sader R, Ghanaati S. Efficacy of platelet-rich fibrin in promoting the healing of extraction sockets: a systematic review. Int J implant dentistry. 2021;7(1):117.
45. La Rosa GRM, Marcianò A, Priolo CY, Peditto M, Pedullà E, Bianchi A. Effectiveness of the platelet-rich fibrin in the control of pain associated with alveolar osteitis: a scoping review. Clin Oral Invest. 2023;27(7):3321–30.
46. Demirok SO, Eroglu CN, Koc A. Comprehensive analysis of bone tissue in extraction sockets of third molars after leukocyte and platelet rich fibrin and photobiomodulation applications. Clin Oral Invest. 2024;28(9):483.
47. Dompe C, Moncrieff L, Matys J, Grzech-Leśniak K, Kocherova I, Bryja A, et al. Photobiomodulation—underlying mechanism and clinical applications. J Clin Med. 2020;9(6):1724.
48. Gholami L, Asefi S, Hooshyarfard A, Sculean A, Romanos GE, Aoki A et al. Photobiomodulation in periodontology and implant dentistry: Part 1. Photobiomodul Photomed Laser Surg. 2019;37(12):739-765. https://doi.org/10.1089/photob.2019.4710.
49. Srinivas B, Das P, Rana MM, Qureshi AQ, Vaidya KC, Raziuddin SJA. Wound healing and bone regeneration in postextraction sockets with and without platelet-rich fibrin. Annals maxillofacial Surg. 2018;8(1):28–34.
50. Azangookhiavi H, Habibzadeh S, Zahmatkesh H, Mellati E, Mosaddad SA, Dadpour Y. The effect of platelet-rich fibrin (PRF) versus freeze-dried bone allograft (FDBA) used in alveolar ridge preservation on the peri-implant soft and hard tissues: a randomized clinical trial. BMC Oral Health. 2024;24(1):693.
51. Leblebicioglu B, Salas M, Ort Y, Johnson A, Yildiz VO, Kim DG, et al. Determinants of alveolar ridge preservation differ by anatomic location. J Clin Periodontol. 2013;40(4):387–95.
52. Freitas I, Baranauskas V, Cruz-Höfling M. Laser effects on osteogenesis. Appl Surf Sci. 2000;154:548–54.
53. Suttapreyasri S, Leepong N. Influence of platelet-rich fibrin on alveolar ridge preservation. J Craniofac Surg. 2013;24(4):1088–94.
54. Thalaimalai DBR, Victor DJ, Prakash PSG, Subramaniam S, Cholan PK. Effect of low-level laser therapy and platelet-rich fibrin on the treatment of intra-bony defects. J Lasers Med Sci. 2020;11(4):456.

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